

Day – 10 Weekly Review and Practice Practical

Task – 1 Make topology of 3 switch and 2 VLANS.

- Switch0
- FastEthernet0/1: Assigned to VLAN 10 (Sales), connected to PC0 (IP: 192.168.10.11)
- FastEthernet0/2: Assigned to VLAN 20 (Support), connected to PC1 (IP: 192.168.20.21)
- FastEthernet0/24: Configured as a Trunk port, connected to Switch1
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- Switch1
- FastEthernet0/1: Assigned to VLAN 10 (Sales), connected to PC2 (IP: 192.168.10.12)
- FastEthernet0/2: Assigned to VLAN 20 (Support), connected to PC3 (IP: 192.168.20.22)
- FastEthernet0/23: Configured as a Trunk port, connected to Switch0
- FastEthernet0/24: Configured as a Trunk port, connected to Switch2
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- Switch2
- FastEthernet0/1: Assigned to VLAN 10 (Sales), connected to PC4 (IP: 192.168.10.13)
- FastEthernet0/2: Assigned to VLAN 20 (Support), connected to PC5 (IP: 192.168.20.23)
- FastEthernet0/24: Configured as a Trunk port, connected to switch 1

Task – 2 Put trunk port between switch to allow traffic of VLANs

- 1. Trunk Port Configuration Commands :- Run the following commands on the interconnecting ports of each switch (for example, FastEthernet0/23 and FastEthernet0/24):
- Switch> enable
- Switch# configure terminal
- Switch(config)# interface range FastEthernet0/23 - 24
- Switch(config-if-range)# switchport mode trunk
- Switch(config-if-range)# end
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- 2. Verification Command :- To check whether the trunking encapsulation has come up properly and which VLANs are allowed across the link, run this command on the switch CLI:
- Switch# show interfaces trunk

- 3. Expected Output to Look For in Your Screenshot :- When you execute show interfaces trunk on a properly configured switch, the output in your Packet Tracer CLI should look similar to this:
- Port Mode Encapsulation Status Native vlan
- Fa0/23 on 802.1q trunking 1
- Fa0/24 on 802.1q trunking 1
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- Port Vlans allowed on trunk
- Fa0/23 1-1005
- Fa0/24 1-1005
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- Port Vlans allowed and active in management domain
- Fa0/23 1,10,20
- Fa0/24 1,10,20
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- Port Vlans in spanning tree forwarding state and not pruned
- Fa0/23 1,10,20
- Fa0/24 1,10,20

Task – 3 Setting up Layer3 Devices.

- 1. Router Configuration (Router-on-a-Stick)
- Connect to the router CLI and execute the following commands:
- Router> enable
- Router# configure terminal
- ! Enable the physical interface
- Router(config)# interface GigabitEthernet0/0
- Router(config-if)# no shutdown
- ! Configure sub-interface for VLAN 10 (Sales)
- Router(config-if)# interface GigabitEthernet0/0.10
- Router(config-subif)# encapsulation dot1q 10
- Router(config-subif)# ip address 192.168.10.1 255.255.255.0
- ! Configure sub-interface for VLAN 20 (Support)
- Router(config-subif)# interface GigabitEthernet0/0.20

- Router(config-subif)# encapsulation dot1q 20
- Router(config-subif)# ip address 192.168.20.1 255.255.255.0
- Router(config-subif)# end
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- 2. Switch Trunk Port Configuration
- Ensure that the switch port connected to the router (GigabitEthernet0/0 on the router linking to Switch0 port Fa0/22) is also configured as a trunk port:
- Switch0> enable
- Switch0# configure terminal
- Switch0(config)# interface FastEthernet0/22
- Switch0(config-if)# switchport mode trunk
- Switch0(config-if)# end
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- 3. Update PC Default Gateways
- To allow the PCs to route traffic across different VLANs, update each PC's Default Gateway in Packet Tracer to match the corresponding router sub-interface IP:
- Sales VLAN (VLAN 10) PCs (PC0, PC2, PC4):
- Subnet: 192.168.10.0/24
- Default Gateway: 192.168.10.1
- Support VLAN (VLAN 20) PCs (PC1, PC3, PC5):
- Subnet: 192.168.20.0/24
- Default Gateway: 192.168.20.1

Task – 4 Testing the connectivity.

- Step 1: Run the Ping Test
- Click on PC0 (Sales - 192.168.10.11), go to the Desktop tab, and open the Command Prompt.
- Type the following command to ping PC1 (Support - 192.168.20.21):
- ping 192.168.20.21
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- Step 2: Troubleshooting Checklist (If Pings Fail)
- If the pings completely fail or result in 100% packet loss, work through these common checks based on your hint:
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- 1. Check PC Default Gateways
- PC0 (Sales): Must have its Default Gateway set to 192.168.10.1.
- PC1 (Support): Must have its Default Gateway set to 192.168.20.1.

- *Fix:* Re-check the IP configuration window on the PCs if traffic isn't leaving the host.
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- 2. Check Router Sub-Interfaces & Encapsulation
- Run show ip interface brief on the router to ensure GigabitEthernet0/0.10 and GigabitEthernet0/0.20 are up/up (and the physical interface GigabitEthernet0/0 has no shutdown).
- Run show running-config to verify that encapsulation dot1q 10 and encapsulation dot1q 20 match their respective sub-interface numbers.
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- 3. Check Switch Trunk Configuration
- Run show interfaces trunk on Switch0 (and any other intermediate switches).
- Ensure the switch port connecting to the router (e.g., FastEthernet0/22) and the inter-switch links (FastEthernet0/23 or FastEthernet0/24) are actively trunking (802.1q). If a port is stuck in access mode, tagged VLAN traffic will be dropped.
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- 4. Check Access Port VLAN Assignments
- Run show vlan brief on all switches.
- Confirm that the ports where PC0 and PC1 are plugged in are assigned strictly to VLAN 10 and VLAN 20 respectively, and not left in the default VLAN 1.
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